ASSIGNMENT 7 / COMPLETE AI DEV PIPELINE

The Suborned

Sky Rumsey · Unity 6000.6.2f1 · Audit scope: live roster portraits, brief 033

01 / Production and functionality

Produced: four agents × nine expression sprites = 36 production portraits. Seedream generates reference-conditioned sheets; OpenCV detects the rings; Python crops, masks and exports consistently named PNGs. The build calls `KitImport.LivePortraits`, which copies and imports sprites automatically. Runtime expressions follow activity, attention, firing and health. All 36 source finals match the tracked Unity assets and local build 0.2.232 inputs.

Manual steps: a person chooses acceptable takes. Four approved downed replacements were separately ring-refined and desaturated; the original grid exporter does not reproduce that finishing. Paid inputs/finals live outside the source-only checkout. Publication and a stranger's first-play check remain manual.

To remove the friction: retain a versioned source-asset archive and selection/override manifest, implement one deterministic downed-state override exporter, then chain export, validation, Unity import, build, signing and upload. Keep aesthetic approval explicit rather than silently treating it as automated.

02 / Architectural reflection

Change: selected generated images and finishing decisions are held in local folders and prose briefs, so a clean checkout cannot reconstruct every portrait. **Alternative:** a content-addressed asset archive plus a manifest containing receipt ID, input hash, selected cell, crop, mask, grade and output hash. One build task consumes it and fails on missing or changed inputs. This makes manual art decisions reproducible data instead of undocumented file edits.

03 / Actual run cost and sustainability

Actual recorded campaign cost: $1.15560 across 14 successful calls, including superseded takes. Nine-state sheets are the most expensive step: 10 calls at $0.0963, totalling $0.9630; four targeted repair strips cost $0.19260. Model: `bytedance:seedream@5.0-pro`, via Runware. These are recorded provider costs, not a forecast.

Asset API cost is affordable for a solo developer at this scale. Review, retries and finishing still cost time; per-pass budgets and reusing approved outputs contain spend. Subscription fees, labour, hardware, hosting and other game assets are excluded. This is the portrait campaign total, not the whole game's production cost.

04 / Mid-project cost reduction

Before: regenerate a 2048×2048 nine-state sheet ($0.0963 per call). **After:** request a 2048×704 strip with three alternatives only for the rejected downed state ($0.04815 per call). The four actual t5 repair calls totalled $0.19260: **50% less per call**. Scope changed; this is not a per-portrait or equal-work benchmark. The approved strip replaces one state while the other eight are retained.

Evidence: `evidence/generation-ledger.json`, `costs.json`, `integration.json`; production scripts under `tools/figures/`; importer and runtime under `unity/Assets/Suborned/`. Receipts were migrated from PROVENANCE; timestamps retain their mtime provenance. Original generation is historical; deterministic replay makes no model calls.

> Submission status: public page offers 0.2.231; local evidence is 0.2.232. Public pipeline video and the ≤120-second cold-start play test remain outstanding.